

SESSION EIGHTEEN CAPSTON PROJECT IN PYTHON

Capstone Project 1: Machine Learning-Based Retail Sales Performance Analysis and Predictive Modeling

ProjectLink: https://drive.google.com/file/d/11U08FO9dDt1gedJeK1jkfX08awqq2xR6/view?usp=drive_link

DatasetLink: https://drive.google.com/file/d/15ITnPtDyA1mr2bkiEwJgor4FZ0ZQ_8Bj/view?usp=drive_link

Project Scenario

Background

ABC Retail Stores Ltd. is a fast-growing retail company operating multiple branches across different regions. The company offers a wide range of consumer products, including electronics, furniture, office supplies, clothing, and household goods. Over the past several years, the organization has accumulated a large transactional database containing customer purchases, pricing information, discounts, inventory records, customer demographics, and sales performance across different locations.

Although management routinely generates descriptive sales reports, these reports only explain historical performance and do not provide predictive insights that can support strategic planning. The company seeks to leverage machine learning techniques to forecast sales, understand customer purchasing behavior, identify factors influencing sales performance, and improve business decision-making.

As a Data Scientist, you have been assigned to develop a complete machine learning pipeline using Python. Your work will involve data preprocessing, exploratory data analysis, feature engineering, predictive modeling, model evaluation, and communicating actionable business insights.

Business Problem

Senior management seeks answers to several strategic questions, including:

- Which factors have the greatest influence on retail sales?
- Can future sales be accurately predicted using historical transaction data?
- Which customer segments generate the highest revenue?
- How do discounts influence sales performance?
- Which regions consistently outperform others?
- Which products experience seasonal demand?
- How can inventory planning be improved using predictive analytics?
- Which stores are likely to achieve higher future sales?
- How can marketing campaigns be optimized based on customer purchasing behavior?

Expected Learning Outcomes

Upon successful completion of the project, students should be able to:

- Develop end-to-end machine learning workflows using Python.
- Prepare and preprocess retail transaction data for analysis.
- Perform exploratory data analysis to identify meaningful patterns.
- Engineer features that improve predictive performance.
- Build and compare multiple regression and classification models.
- Evaluate models using appropriate statistical and machine learning metrics.
- Interpret model outputs and feature importance.
- Communicate analytical findings through professional reports and presentations.
- Apply best practices in reproducible data science, including code organization, documentation, and version control.

NB: This capstone project mirrors a real-world data science engagement in the retail sector, providing students with practical experience in transforming raw transactional data into actionable predictive insights using Python and standard machine learning techniques.

Capstone Project 2: Machine Learning-Based Agricultural Crop Production Analysis and Yield Prediction

ProjectLink: https://drive.google.com/file/d/1amapXFLHwclSUgdConiwxqKzxKLKmxh9/view?usp=drive_link

DatasetLink: https://drive.google.com/file/d/1Nlj794dW7bSzLeNKr5AS0mKNfSYFbsfO/view?usp=drive_link

Project Scenario

Background Green Harvest Farmers' Cooperative Society is a national agricultural organization that supports thousands of farmers across different agricultural regions. The cooperative collects production data for major crops, including maize, wheat, rice, beans, potatoes, and sorghum. The data includes information on farm characteristics, weather conditions, farming practices, fertilizer application, irrigation methods, labor, and market prices.

Although the cooperative has accumulated several years of agricultural production records, it currently relies on descriptive reports that only summarize historical performance. Management would like to leverage machine learning to predict crop production, identify the factors influencing yield, and support evidence-based agricultural planning.

As a Data Scientist, you have been assigned to develop an end-to-end machine learning solution using Python. Your work will involve data preprocessing, exploratory data analysis, feature engineering, predictive modeling, model evaluation, and communicating actionable insights that can improve agricultural productivity

Business Problem The cooperative seeks answers to the following questions:

- Which factors have the greatest impact on crop production?
 - Can crop yield be accurately predicted before harvest?
 - Which farming regions consistently produce the highest yields?
 - How do rainfall, fertilizer usage, and irrigation influence production?
 - Which crops are most profitable?
 - How does farm size affect production?
 - Which farming practices contribute to higher yields?
 - How can the cooperative improve resource allocation and production planning?
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Expected Learning Outcomes

Upon successful completion, students should be able to:

- Develop an end-to-end machine learning workflow using Python.
- Prepare agricultural datasets for predictive modeling.
- Perform exploratory data analysis and feature engineering.
- Build and compare multiple regression and classification models.
- Evaluate model performance using standard machine learning metrics.
- Interpret model outputs and feature importance.
- Communicate analytical findings and practical recommendations to stakeholders.

Capstone Project 3: Machine Learning-Based Customer Banking Transactions Analysis and

Predictive Modeling

ProjectLink: https://drive.google.com/file/d/1VxI0EsVNbudW4PhnJi7leJP-8Pwqliad/view?usp=drive_link

DatasetLink: https://drive.google.com/file/d/1gOeu49H9LINgA1EkUqClhCC2iF3vahO5/view?usp=drive_link

Project Scenario

Background SecureBank Ltd. is a rapidly growing commercial bank with branches across multiple regions. The bank serves individual and corporate customers through branch banking, ATMs, internet banking, mobile banking, and agency banking services. Every day, thousands of financial transactions are processed and stored in the bank's centralized database.

Over the years, the bank has accumulated a substantial volume of customer transaction data containing information on customer demographics, account details, transaction types, transaction channels, balances, and banking activities. Although management regularly reviews descriptive reports, the bank has not fully utilized its data to develop predictive models that can support operational planning and customer relationship management.

The management has commissioned a data science project to analyze customer transaction data and develop machine learning models capable of identifying customer transaction patterns and predicting future banking behavior.

As a Data Scientist, your responsibility is to develop a complete machine learning workflow using Python, including data preprocessing, exploratory data analysis, feature engineering, predictive modeling, model evaluation, and interpretation of results.

Business Problem

The bank seeks to answer the following strategic questions:

- Which customer characteristics influence transaction behavior?
 - Can the bank predict whether a customer is likely to become a high-value customer?
 - Which transaction channels are associated with higher transaction volumes?
 - Which factors influence account balances?
 - Which customer segments generate the highest banking activity?
 - Can transaction amounts be accurately predicted?
 - Which branches consistently record higher transaction volumes?
 - Which banking products are most associated with customer retention?
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Expected Learning Outcomes

Upon successful completion of this capstone project, students will be able to:

- Develop an end-to-end machine learning workflow using Python.
- Prepare and preprocess banking transaction data for predictive modeling.
- Perform exploratory data analysis and feature engineering.
- Build and compare multiple regression and classification models.
- Evaluate model performance using standard machine learning metrics.
- Interpret feature importance and model outputs.
- Present actionable business insights and recommendations based on predictive analytics.
- Apply reproducible data science practices, including well-documented code, structured reporting, and effective presentation of results.